Empirical Methods for Policy Evaluation

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Part 3: Difference-in-Differences and Job Search Models

- Two-way fixed effect regressions
  - $\Rightarrow$ DID $\neq$ TWFE
  - $\Rightarrow$ Event Study regressions
  - $\Rightarrow$ Heterogeneity-robust DID estimators
- Empirical job search models
  - $\Rightarrow$ Eckstein-van den Berg (JoE, 2007)
- Informality, job search, and schooling investments
  - $\Rightarrow$ Bobba-Flabbi-Levy (IER 2022)

DID $\neq$ TWFE

Groups and Time Periods

- We consider observations that can be divided into G groups and T periods
- For every $(g,t) \in \{1,...,G\} \times \{1,...,T\}$: $=$ nb of obs in group g at period t
  - $\Rightarrow$ For simplicity, we assume hereafter balanced panel of groups:

$$\text{For all } (g,t) \in \{1,...,G\} \times \{1,...,T\}, N_{g,t} > 0$$

- Panel/repeated cross-section data set where groups are, e.g.
  - $\Rightarrow$ Individuals, firms, counties, etc.
  - $\Rightarrow$ Cross-section dataset where cohort of birth plays the role of time
  - $\Rightarrow$ One may have $N_{g,t} = 1$, e.g. b/c group=individual or a firm

Potential Outcomes, SUTVA, and Covariates

- Let $(d_1, ..., d_T)$ denote a treatment trajectory
- Corresponding potential outcomes: $Y_{g,t}(d_1, ..., d_T)$
- Then observed outcome: $Y_{g,t} = Y_{g,t}(D_{g,1}, ..., D_{g,T})$
- We maintain the usual SUTVA:

$$(Y_{g,1}(d_1, ..., d_T), ..., Y_{g,T}(d_1, ..., d_T)) \perp (D_{g',t'})_{g' \neq g, t'=1,...,T}, \forall (g, t, d_1, ..., d_T)$$

- For any scalar $X_{g,t}$, let $\boldsymbol{X}_g = (X_{g,1}, ..., X_{g,T})$ and $\boldsymbol{X} = (\boldsymbol{X}_1, ..., \boldsymbol{X}_G)$

The Simplest Setup, 2x2 Case

- Two groups $g \in \{s, n\}$ and two time periods $t \in \{1, 2\}$
- $D_{g,t} \in \{0, 1\}$, such that $D_{s,1} = D_{n,1} = 0$, $D_{s,2} = 1$, and $D_{n,2} = 0$
- $Y_{g,t}$ is the observed outcome in location g at period t
- $Y_{g,t}(0), Y_{g,t}(1)$ are potential outcomes without and with treatment

The Parallel (//) Trend Assumption

- In the absence of treatment, same average outcome evolution across groups

$$\mathbb{E}[Y_{s,2}(0) - Y_{s,1}(0)] = \mathbb{E}[Y_{n,2}(0) - Y_{n,1}(0)]$$

- Weaker than imposing that s and n have same untreated-outcome levels

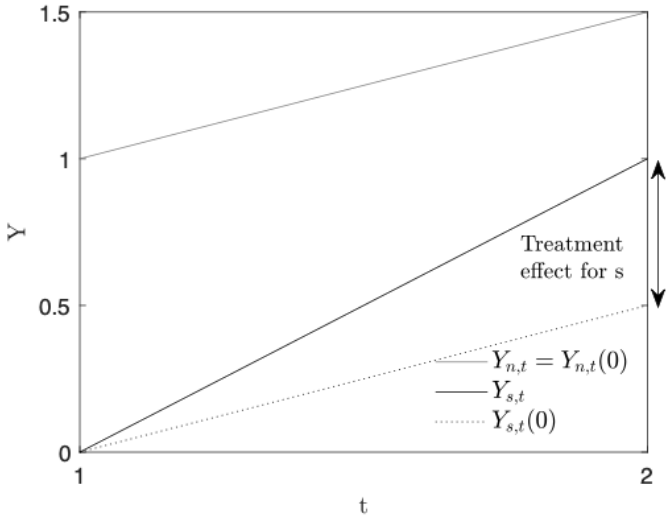
$$\mathbb{E}[Y_{s,t}(0)] = \mathbb{E}[Y_{n,t}(0)] \text{ for all } t$$

- Also weaker than imposing no variation in average untreated outcomes

$$\mathbb{E}[Y_{g,2}(0)] = \mathbb{E}[Y_{g,1}(0)] \text{ for all } g$$

- Appeal of // trends: has testable implications (no pre-trends)

Illustration of Parallel-trends



In the Simplest Set-up, TWFE = DID

- Under // trends, DID is unbiased for the ATE in location s at period 2

$$\begin{aligned}\mathbb{E}(DID) &= \mathbb{E}[Y_{s,2} - Y_{s,1} - (Y_{n,2} - Y_{n,1})] \\ &= \mathbb{E}[Y_{s,2}(1) - Y_{s,1}(0) - (Y_{n,2}(0) - Y_{n,1}(0))] \\ &= \mathbb{E}[Y_{s,2}(1) - Y_{s,2}(0)] + \mathbb{E}[Y_{s,2}(0) - Y_{s,1}(0)] - \mathbb{E}[Y_{n,2}(0) - Y_{n,1}(0)] \\ &= \mathbb{E}[Y_{s,2}(1) - Y_{s,2}(0)]\end{aligned}$$

- The before-after diff is combined with the treated-control diff

- This result extends beyond the 2x2 case as long as

  $\Rightarrow$ Absorbing and binary treatment

  $\Rightarrow$ No variation in treatment timing across groups (no staggered design)

The Pervasiveness of Two-way Fixed Effect Regressions

- Researchers often consider two-way fixed effects (TWFE) models of the kind:

$$Y_{g,t} = \alpha_g + \gamma_t + \beta_{fe} D_{g,t} + \epsilon_{g,t}.$$

- 26 out of the 100 most cited 2015-2019 AER papers estimate TWFE
- Also commonly used in other social sciences
- Other popular method: event-study regressions=dynamic version of TWFE

TWFE Parameters

- $\widehat{\beta}_{fe}$= OLS coeff. of $D_{g,t}$ in a reg. of $Y_{g,t}$ on group FEs, time FEs and $D_{g,t}$
- $\widehat{\beta}_{fd}$= OLS in a reg. of $Y_{g,t} - Y_{g,t-1}$ on time FEs and $D_{g,t} - D_{g,t-1}$
- $\widehat{\beta}^{X}_{fe}$, when include covariates $X_{g,t}$ in the regression
- We first focus on β_{fe}, but we will extend the results to β_{fd} and β^{X}_{fe}

Additive Separability of TWFE

- A more general, yet static, case

$$D_{g,t} \in R^{+} \text{and for all } (g,t,d_1,...,d_T), Y_{g,t}(d_1,...,d_T) = Y_{g,t}(d_t)$$

- Parallel trends: for all $t \geq 2$, $E[Y_{g,t}(0) - Y_{g,t-1}(0)] = \gamma_t$
- It follows that: $E[Y_{g,t}(0) - Y_{g,1}(0)] = \gamma_t$, and let $\alpha_g = E[Y_{g,1}(0)]$
- Then, $E[Y_{g,t}(0)] = E[Y_{g,1}(0)] + E[Y_{g,t}(0) - Y_{g,1}(0)] = \alpha_g + \gamma_t$

Parameter of Interest

- Average treatment response

$$\Delta^{TR} = \frac{1}{\sum_{g,t} D_{g,t}} \sum_{g,t} \left(Y_{g,t}(D_{g,t}) - Y_{g,t}(0) \right)$$

- Then, let $\delta^{TR} = E[\Delta^{TR}]$. With a binary D, δ^{TR} =ATT

- Analogously, in (g,t):

$$\Delta_{g,t} = \frac{1}{D_{g,t}} \left[Y_{g,t}(D_{g,t}) - Y_{g,t}(0) \right] \text{ if } D_{g,t} \neq 0$$

- Then:

$$\delta^{TR} = E\left[\sum_{(g,t):D_{g,t}>0} W_{g,t} \Delta_{g,t} \right], \quad \text{with } W_{g,t} = \frac{D_{g,t}}{\sum_{(g,t):D_{g,t}>0} D_{g,t}}$$

β_{fe} = weighted sum of ATEs under // trends

- Let $\epsilon_{fe,g,t}$= resid. of the reg. of $D_{g,t}$ on a constant, group FEs, and time FEs
- It can be shown that:

$$\beta_{fe} = E\left[\sum_{(g,t):D_{g,t}>0} W_{fe,g,t}\Delta_{g,t}\right]$$

$\Rightarrow W_{fe,g,t} = \frac{D_{g,t}\epsilon_{fe,g,t}}{\sum_{(g,t):D_{g,t}>0} D_{g,t}\epsilon_{fe,g,t}}$

- In general, $\beta_{fe} \neq \delta^{TR}$ because $W_{fe,g,t} \neq W_{g,t}$
  - $\Rightarrow$ We may have $W_{fe,g,t} < 0$: if $\epsilon_{fe,g,t} < 0$ while $D_{g,t} > 0$
  - $\Rightarrow$ $E\left[\widehat{\beta}_{fe}\right]$ may be < 0 even if $Y_{g,t}(d) > Y_{g,t}(0)$ for all (g,t) and $d > 0$

Characterizing (g, t) cells weighted negatively by β_{fe}

- Let $D_{g,\cdot}$ =average treat. rate of g and $D_{\cdot,t}$ =average treat. rate at t
- Under // trends, $W_{fe,g,t}$ is decreasing with $D_{g,\cdot}$ and $D_{\cdot,t}$
- β_{fe} more likely to assign negative weight to:
  - $\Rightarrow$ Periods with many vs few treated groups
  - $\Rightarrow$ Groups treated for many vs few periods
- In staggered adoption designs ($D_{g,t} \geq D_{g,t-1}$), $W_{fe,g,t} < 0$ more likely
  - $\Rightarrow$ In the last periods and for groups adopting the treatment earlier
  - $\Rightarrow$ Remove negative weights by removing always treated groups

Forbidden Comparison 1: Switchers Vs. Always Treated

- D binary and design staggered: $\widehat{\beta}_{fe}$ = weighted avg of two types of DIDs:
  - $\Rightarrow$ DID_1: comparing s from untreated to treated to n untreated at both dates
  - $\Rightarrow$ DID_2: comparing switching group s to group a treated at both dates.
- Negative weights in β_{fe} originate from the second type of DIDs

Forbidden Comparison 1: An Example

- Group e treated at $t = 2$, group ℓ treated at $t = 3$. Then:

$$\widehat{\beta}_{fe} = \frac{1}{2} \times \underbrace{DID_{e-\ell}^{1-2}}_{DID_1} + \frac{1}{2} \times \underbrace{DID_{\ell-e}^{2-3}}_{DID_2}$$

- At periods 2 and 3, e's outcome = treated potential outcome, so

$$Y_{e,3} - Y_{e,2} = Y_{e,3}(1) - Y_{e,2}(1) = Y_{e,3}(0) + \Delta_{e,3} - (Y_{e,2}(0) + \Delta_{e,2}).$$

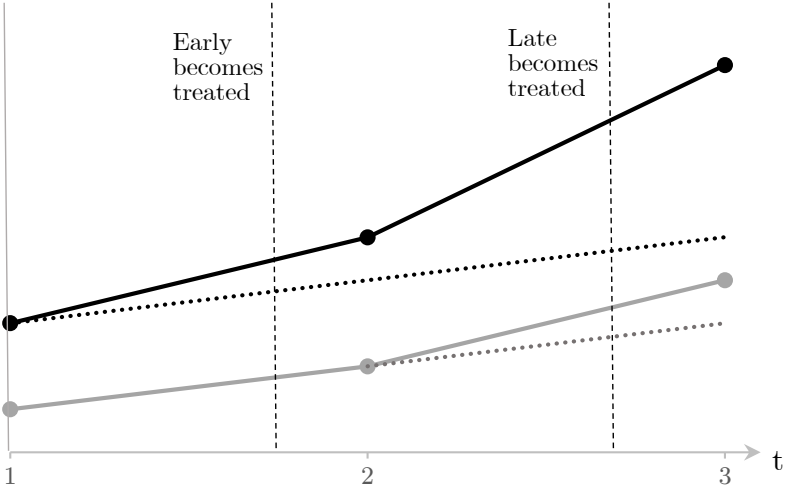
- On the other hand, group ℓ only treated at period 3, so

$$Y_{\ell,3} - Y_{\ell,2} = Y_{\ell,3}(0) + \Delta_{\ell,3} - Y_{\ell,2}(0)$$

Forbidden Comparison 1: An Example (continued)

- $E\left[DID_{\ell-e}^{2-3}\right] = E\left[Y_{\ell,3} - Y_{\ell,2} - (Y_{e,3} - Y_{e,2})\right] = E\left[\Delta_{\ell,3} + \Delta_{e,2} - \Delta_{e,3}\right]$
- Note: if $\Delta_{e,2} = \Delta_{e,3}$, $E[DID_{\ell-e}^{2-3}] = E\left[\Delta_{\ell,3}\right]$
- More generally, if $\Delta_{g,t} = \Delta_{g,t'}$, $W_{fe,g,t} \geq 0$. But restrictive!
- Note:
$$Y_{g,t}(0) - Y_{g,t-1}(0) = Y_{g,t}(1) - Y_{g,t-1}(1) \iff \Delta_{g,t} = \Delta_{g,t-1}$$
- This assumption is actually equivalent to time-invariant treatment effects

Forbidden Comparison 1: Graphical Illustration



Forbidden Comparison 2: Switching more vs Switching less

- Suppose the treatment D is not binary and treatment is not staggered
- $\widehat{\beta}_{fe}=$ (group m whose D increases more) - (group ℓ whose D increases less)
- With two groups m and ℓ and two periods:

$$\widehat{\beta}_{fe} = \frac{Y_{m,2} - Y_{m,1} - (Y_{\ell,2} - Y_{\ell,1})}{D_{m,2} - D_{m,1} - (D_{\ell,2} - D_{\ell,1})}$$

- This "Wald-DID" estimator may not estimate convex combination of effects

Forbidden Comparison 2: An Example

- Assume m goes from 0 to 2 units of treatment while ℓ goes from 0 to 1

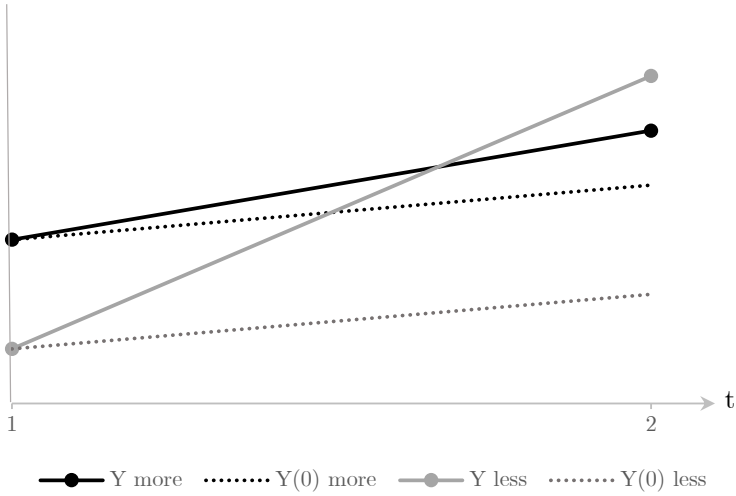
  $\Rightarrow$ Denominator of the Wald-DID is $2-0-(1-0)=1$

- Potential outcomes: $Y_{m,t}(d) = Y_{m,t}(0) + \delta_m d$ and $Y_{\ell,t}(d) = Y_{m,t}(0) + \delta_\ell d$

- Then:

$$\begin{aligned} E\left[\widehat{\beta}_{fe}\right] =& E\left[Y_{m,2} - Y_{m,1} - (Y_{\ell,2} - Y_{\ell,1})\right] \\ =& E\left[Y_{m,2}(0) + 2\delta_m - Y_{m,1}(0) - (Y_{\ell,2}(0) + \delta_\ell - Y_{\ell,1}(0))\right] \\ =& E\left[Y_{m,2}(0) - Y_{m,1}(0)\right] - E\left[Y_{\ell,2}(0) - Y_{\ell,1}(0)\right] + 2\delta_m - \delta_\ell \\ =& 2\delta_m - \delta_l \end{aligned}$$

Forbidden Comparison 2: Graphical Illustration



Extensions

- This logic extends to β_{fd}, but with different weights $W_{fd,g,t}$

  $\Rightarrow$ If $T = 2$, then $\beta_{fd} = \beta_{fe}$

- With covariates, we modify // trends as follows:

$$E[Y_{g,t}(0) - Y_{g,t-1}(0)|\boldsymbol{X}_g] = E[Y_{g',t}(0) - Y_{g',t-1}(0)|\boldsymbol{X}_{g'}], \ \forall g, g' \in \mathcal{G}$$

- Let $\epsilon^X_{fe,g,t}$= resid. of reg. of $D_{g,t}$ on a const., group FEs, time FEs and $X_{g,t}$

- Then, same result as above but with $\epsilon^X_{fe,g,t}$ instead of $\epsilon_{fe,g,t}$ in $W_{fe,g,t}$.

Example: Effect of Newspapers on Electoral Turnout

- Gentzkow et al. (AER, 2011) use US county data on presidential elections
- Reg change in turnout in g on change in # newspapers + state-year FE
- One could also estimate the FE regression

Regression	$\widehat{\beta}$ (s.e.)	% of < 0 weights	Sum of < 0 weights
$\widehat{\beta}_{fe}$	-0.0011 (0.0011)	40.1%	-0.53
$\widehat{\beta}_{fd}$	0.0026 (0.0009)	45.7%	-1.43

$\Rightarrow$ Under // trends, $\beta_{fe} \neq \beta_{fd} \rightarrow \Delta_{g,t} = \Delta \ \forall (g,t)$

Event Study Regressions

Main Assumptions

- Assuming past treatments do not affect current outcomes is restrictive
- Dynamic framework with no anticipation, $D_{g,t} \in R^+$:

$$Y_{g,t}(d_1, ..., d_T) = Y_{g,t}(d_1, ..., d_t)$$

- // trends, v2 ($\mathbf{0}_t$ =vector of t zeros)
  - $\Rightarrow$ $\forall g, g'$ and $t \geq 2$, $E[Y_{g,t}(\mathbf{0}_t) - Y_{g,t-1}(\mathbf{0}_{t-1})] = E[Y_{g',t}(\mathbf{0}_t) - Y_{g',t-1}(\mathbf{0}_{t-1})]$
  - $\Rightarrow$ For all $t \geq 2$, $E[Y_{g,t}(\mathbf{0}_t) - Y_{g,t-1}(\mathbf{0}_{t-1})] = \gamma_t$

Notation and Parameters

- Staggered treatment: $D_{g,t} = 1\{t \geq F_g\}, F_g$ =period when g becomes treated
- Exclude always treated here and let $F_g = T + 1$ if never treated
- Let $\mathbf{1}_t$ be a vector of t ones and define:

$$\Delta_g(\ell) = Y_{g,F_g+\ell}(\mathbf{0}_{F_g-1}, \mathbf{1}_{\ell+1}) - Y_{g,F_g+\ell}(\mathbf{0}_{F_g+\ell}),$$
$$\Delta(\ell) = \frac{1}{N_{\leq}(T-\ell)} \sum_{g:F_g+\ell \leq T} \Delta_g(\ell)$$

$\Rightarrow$ $\Delta_g(\ell)$: effect for g of having received the treatment for $\ell + 1$ periods

$\Rightarrow$ $\Delta(\ell)$: effect of having received the treatment for $\ell + 1$ periods

Event-Study (ES) Regressions

- To estimate the $\Delta(\ell)$, researchers often perform ES regressions of the form

$$Y_{g,t} = \gamma_g + \gamma_t + \sum_{\ell=-K, \ell \neq 1}^{L} \beta_\ell 1\{F_g = t - \ell\} + \epsilon_{g,t}$$

- $1\{F_g = t - \ell\} = 1$ if group g started receiving the treatment ℓ periods ago
- For $\ell \geq 0$, β_ℓ is supposed to estimate $\Delta(\ell)$
- For $\ell \leq -2$, β_ℓ = placebo used to check // trends

  $\Rightarrow$ Outcome trends of groups that will/will not start treatment in $|\ell|$ periods

An Aside: "Basic" Problems with ES Regressions

- Fully-dynamic specification with first and last indicators
  - $\Rightarrow$ $1\{F_g = t + K\}$ and $1\{F_g = t - L\}$
- If all groups are eventually treated, then some β_ℓs not identified
  - $\Rightarrow$ Need never treated groups
  - $\Rightarrow$ Otw some placebo coeffs should be removed, or dynamic effects be restricted

Decomposition Result for ES Regressions

- $\epsilon_{g,t}$: res. of $1\{F_g = t - \ell\}$ on group/time FEs and $(1\{F_g = t - \ell'\})_{\ell'=-K,..,\overline{L}}$
- Define $W_{g,\ell,\ell'} = \frac{\epsilon_{g,F_g+\ell'}}{\sum_g \epsilon_{g,F_g+\ell}}, \forall(\ell,\ell')$
- Then, for $\ell = 0, ..., \overline{L}$,

$$E\left[\widehat{\beta}_\ell\right] = E\Bigg[\underbrace{\sum_g W_{g,\ell,\ell}\Delta_g(\ell)}_{\text{sum of effect of } \ell+1 \text{ treat. periods}} + \underbrace{\sum_{\ell'\neq\ell}\sum_g W_{g,\ell,\ell'}\Delta_g(\ell')}_{\text{sum of effects of } \ell'+1 \text{ treat periods}}\Bigg]$$

ES Even Less Robust to Heterog. TE than Static TWFE

- As in the static case, $W_{g,\ell,\ell}$ may be < 0
- $\Delta_g(\ell) = \Delta_{g'}(\ell)\ \forall g, g'$ not sufficient for $\widehat{\beta}_\ell$ to be unbiased. We also need

$$\Delta_g(\ell') = \Delta_{g'}(\ell')\ \forall \ell' \neq \ell$$

  $\Rightarrow$ $\widehat{\beta}_\ell$ may be contaminated by effects of $\ell' + 1$ treatment periods

- For placebo tests ($\ell \leq -2$) need a stronger assumption

$$\Delta_g(\ell') = \Delta(\ell')\ \forall g, \ell'$$

Origin: Again, Forbidden Comparisons

- As in the static case, $\widehat{\beta}_\ell$ can still be written as a linear combination of DID
- But in some of these DIDs:
  - $\Rightarrow$ The control group has been treated for $\ell' \neq \ell$ period at baseline/endline
  - $\Rightarrow$ If $\ell > 0$, the treated group has been treated for $\ell' > 0$ period at baseline
  - $\Rightarrow$ The treated group has been treated for $\ell' < \ell$ period at endline
- Contamination bias

A Simple Example

- Consider a design with $G = 2$, $T = 3$, $K = 0$ (no placebo tests)
  - Group 1 treated at $t = 2$
  - Group 2 treated at $t = 3$

- Then, some algebra shows that:

$$\widehat{\beta}_1 = \underbrace{Y_{1,3} - Y_{1,2} - (Y_{2,3} - Y_{2,2})}_{DID_1} + 2\underbrace{[Y_{1,2} - Y_{1,1} - (Y_{2,2} - Y_{2,1})]}_{DID_2}$$

Example Continued

- DID_1: g=2 treated for 1 period at t=3 and g=1 treated for 1 period at t=2

$$E[DID_1] = E[\Delta_1(1) - (\Delta_1(0) + \Delta_2(0))]$$

- DID_2 identifies the effect of having been treated for 1 rather than 2 periods:

$$E[DID_2] = E[\Delta_1(0)]$$

- We obtain $E[\widehat{\beta}_1] = E[\Delta_1(1) + (\Delta_1(0) - \Delta_2(0))]$

Heterogeneity-robust DID estimators

Robust DIDs (Static Case)

- Avoid making the forbidden comparisons leveraged by TWFE:
  - $\Rightarrow$ Never compare switcher to switcher: only compare switcher to stayer
  - $\Rightarrow$ Never compare a switcher to a stayer with a different baseline treatment
- In static case, consider each pair of consecutive time periods independently
  - $\Rightarrow$ $t-1$ to t switchers: groups whose treatment changes from $t-1$ to t
  - $\Rightarrow$ $t-1$ to t stayers: groups whose treatment does not change from $t-1$ to t, with same $t-1$ treatment as switchers

Robust DIDs (Dynamic Case)

- In more general case, we need to control for groups' full treatment history
  - $\Rightarrow$ $t-1$ to t first-time switchers: treat changes for the first time from $t-1$ to t
  - $\Rightarrow$ 1 to $t+\ell$ stayers: treat does not change from 1 to $t+\ell$, with same $t-1$ treatment as switchers
- Allowing for dynamic effects may lead to less precise and interpretable effects

Parameters of interest (Static Case)

- Binary staggered design: $D_{g,t} = 1\{t \geq F_g\}$
- Let us define

$$\mathcal{S} = \{(g,t) : t \geq 2,\ D_{g,t} \neq D_{g,t-1},\ \exists g' :\ D_{g',t} = D_{g',t-1} = D_{g,t-1}\}$$

- N_S =card($\mathcal{S}$)
- ATE for matchable switchers is

$$\delta^S = E\left[\frac{1}{N_S}\sum_{(g,t)\in\mathcal{S}} Y_{g,t}(1) - Y_{g,t}(0)\right]$$

Assumptions for identifying δ^S

- δ^S can be unbiasedly estimated under the following // trends conditions

$$E[Y_{g,t}(0) - Y_{g,t-1}(0)|\boldsymbol{D}_g] = E[Y_{g,t}(0) - Y_{g,t-1}(0)] = \gamma_{0,t}$$
$$E[Y_{g,t}(1) - Y_{g,t-1}(1)|\boldsymbol{D}_g] = E[Y_{g,t}(1) - Y_{g,t-1}(1)] = \gamma_{1,t}$$

- Usual // trends on $Y_{g,t}(0)$ sufficient if we focus on switchers in

$$\mathcal{S}_+ = \{(g,t) : t \geq 2,\ D_{g,t} = 1 > D_{g,t-1} = 0,\ \exists g' :\ D_{g',t} = D_{g',t-1} = 0\}$$

Weighted averages of DIDs identify δ^S

- For all $t \in \{1, ..., T\}$ and $d = 0, 1$, let

$$E[DIDM] = E\left[\sum_{t=2}^{T} \frac{N_{+,t}}{N_S} DID_{+,t} + \frac{N_{-,t}}{N_S} DID_{-,t}\right] = \delta^S$$

$\Rightarrow N_{+,t} = \text{card}\{g : D_{g,t} > D_{g,t-1}\}; N_{-,t} = \text{card}\{g : D_{g,t} < D_{g,t-1}\}; N_{=d,t} = \text{card}\{g : D_{g,t} = D_{g,t-1}\}$

$\Rightarrow DID_{+,t} = \sum_{g:D_{g,t}>D_{g,t-1}} \frac{1}{N_{+,t}} (Y_{g,t} - Y_{g,t-1}) - \sum_{g:D_{g,t}=D_{g,t-1}=0} \frac{1}{N_{=0,t}} (Y_{g,t} - Y_{g,t-1})$

$\Rightarrow DID_{-,t} = \sum_{g:D_{g,t}=D_{g,t-1}=1} \frac{1}{N_{=1,t}} (Y_{g,t} - Y_{g,t-1}) - \sum_{g:D_{g,t}<D_{g,t-1}} \frac{1}{N_{-,t}} (Y_{g,t} - Y_{g,t-1})$

- Placebo estimators

$\Rightarrow$ Focus on groups that are stayers one, two, three, etc. periods before switchers switch

Example (continued): Gentzkov et al. AER, 2011

	Estimate	Standard error	N
$\widehat{\beta}_{fd}$	0.0026	0.0009	15,627
$\widehat{\beta}_{fe}$	-0.0011	0.0011	16,872
$DIDM$	0.0043	0.0014	16,872
$DIDM$ Placebo	-0.0009	0.0016	13,221

- $DIDM$ is 66% larger (t-stat=1.77) than $\widehat{\beta}_{fd}$ and opposite sign to $\widehat{\beta}_{fe}$

Allowing for Dynamic Effects

- We maintain previous assumption of no anticipation effects

  $\Rightarrow$ Potential outcomes $Y_{g,t}(d_1, ..., d_t)$ do not depend on future treatments

- F_g = date at which group g starts receiving the treatment

- $\mathbf{0}_k$: vector of k zeros, $\mathbf{1}_k$: vector of k ones

- $Y_{g,t}(\mathbf{0}_k, \mathbf{1}_{t-k})$: g at t if untreated from 1 to k and treated from $k+1$ to t

Parameters of Interest

- ATE of cohort getting treated from c to $c+\ell$, i.e. for $\ell+1$ periods

$$\delta(c,\ell) = E\left[\frac{1}{N_{=}(c)}\sum_{g:F_g=c}\left(Y_{g,c+\ell}(\mathbf{0}_{c-1},\mathbf{1}_{\ell+1}) - Y_{g,c+\ell}(\mathbf{0}_{c+\ell})\right)\middle|\boldsymbol{D}\right]$$

- Where $N_{=}(c) = \text{card}\{g : F_g = c\}$

- ATE across cohorts treated for $\ell+1$ periods

$$\delta(\ell) = E\left[\frac{1}{\sum_{c=2}^{T-\ell} N_{=}(c)}\sum_{c=2}^{T-\ell} N_{=}(c)\delta(c,\ell)\middle|\boldsymbol{D}\right]$$

DID Estimators Using Never-treated Groups

- To estimate $\delta(c, \ell)$

$$\widehat{\delta}_1(c,\ell) = \frac{1}{N_=(c)} \sum_{g:F_g=c} (Y_{g,c+\ell} - Y_{g,c-1}) - \frac{1}{N_=(T+1)} \sum_{g:F_g=T+1} (Y_{g,c+\ell} - Y_{g,c-1})$$

- Recall that $\{g : F_g = T+1\}$ = never-treated groups

- $\widehat{\delta}_1(c,\ell)$: ΔY between $c-1$ and $c+\ell$ in cohort c and in never-treated groups

- Then let

$$\widehat{\delta}_1(\ell) = \frac{1}{\sum_{c=2}^{T-\ell} N_=(c)} \sum_{c=2}^{T-\ell} N_=(c) \widehat{\delta}_1(c,\ell)$$

DID Estimators Using Other Control Groups

- Never-treated groups may not exist or be different from other groups $\rightarrow$ // trends may not hold
- There are more not-yet-treated groups than never-treated groups

$$\widehat{\delta}_2(c,\ell) = \frac{1}{N_=(c)} \sum_{g:F_g=c} (Y_{g,c+\ell} - Y_{g,c-1}) - \frac{1}{N_>(c+\ell)} \sum_{g:F_g>c+\ell} (Y_{g,c+\ell} - Y_{g,c-1})$$

  $\Rightarrow$ $\widehat{\delta}_2(c,l)$ possibly more precise than $\widehat{\delta}_1(c,\ell)$

- $N_>(c+\ell) = \text{card}\{g : F_g > c+\ell\}$, and define $\widehat{\delta}_2(\ell)$ accordingly
- Define placebo estimators accordingly

The Simplest Job Search Model

Optimal Stopping

- Risk neutral individual in discrete time with preferences in $t = 0$ given by

$$\sum_{t=0}^{\infty} \beta^t c_t, \quad \beta \in (0, 1)$$

- Start as unemployed, with consumption equal to b
- Jobs sampled sequentially; each job is for life and identical except for wage
- Wages are drawn from an exogenous stationary distribution $F(w)$
- Given draw $w_t \in W$ agent decides whether to take it ($c_t = w_t$) or continue searching

Dynamic Programming Formulation

- Value function for the agent when he has sampled a job of $w \in W$ is

$$V(w) = \max\left\{\frac{w}{1-\beta}, \beta V + b\right\}$$

- V is the continuation value of not accepting a job:

$$V = \int_{\omega\in\Omega} V(\omega) dF(\omega)$$

- Combine these two equations and get:

$$V(w) = \max\left\{\frac{w}{1-\beta}, b + \beta \int_{\omega\in\Omega} V(\omega) dF(\omega)\right\}$$

Reservation Wage

- $V(w)$ is non-decreasing, decision rule has a reservation value property

$$\frac{w^\star}{1-\beta} = b + \beta \int_{\omega \in \Omega} V(\omega) dF(\omega)$$

- Decision rule: $\forall w < w^\star$, $V(w) = \frac{w^\star}{1-\beta}$ and $\forall w \geq w^\star$, $V(w) = \frac{w}{1-\beta}$

- Therefore, reservation wage can be written as

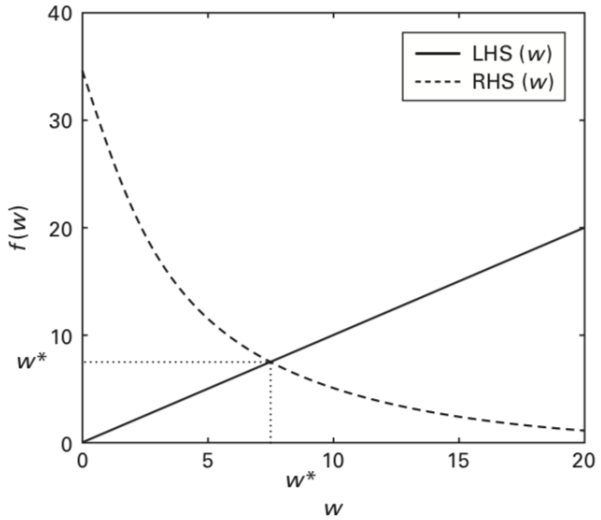
$$\frac{w^\star}{1-\beta} = b + \beta \left[\frac{w^\star F(w^\star)}{1-\beta} + \int_{w \geq w^\star} \frac{w}{1-\beta} dF(w) \right]$$

Reservation Wage

- Since $\frac{w^\star}{1-\beta} = \int_{w<w^\star} \frac{w^\star}{1-\beta} dF(w) + \int_{w \geq w^\star} \frac{w^\star}{1-\beta} dF(w)$
- Subtract from both sides: $\int_{w<w^\star} \beta \frac{w^\star}{1-\beta} dF(w) + \int_{w \geq w^\star} \beta \frac{w^\star}{1-\beta} dF(w)$
- And you get

$$w^\star = b + \frac{\beta}{1-\beta} \left[\int_{w \geq w^\star} (w - w^\star) dF(w) \right]$$

Equilibrium



Taking the Model to the Data

- For a random sample of N workers we observe $\{\tilde{t}_u(i), w(i)\}_{i=1}^N$
- Job offers/termination arrive at random times with density between offers

$$\begin{aligned} q_u(t_u) &= \lambda \exp(-\lambda t_u), \lambda > 0 \\ q_e(t_e) &= \eta \exp(-\eta t_e), \eta > 0 \end{aligned}$$

- Reservation wage in continuous time

$$w^\star = b + \frac{\lambda}{\rho + \eta} \int_{w^\star} (w - w^\star) dF(w)$$

$\Rightarrow$ Easy to show that $\partial w^\star / \partial \eta < 0, \partial w^\star / \partial \rho < 0, \partial w^\star / \partial \lambda > 0, w^\star / \partial(\mu_w, \sigma_w) > 0$

Steady-state Proportions

- The probability that an individual is unemployed is

$$p(u) = \frac{\mathbb{E}(t_u)}{\mathbb{E}(t_e) + \mathbb{E}(t_u)} = \frac{[\lambda \tilde{F}(w^\star)]^{-1}}{\eta^{-1} + [\lambda \tilde{F}(w^\star)]^{-1}} = \frac{\eta}{\eta + \lambda \tilde{F}(w^\star)}$$

- And conversely

$$p(e) = 1 - p(u) = \frac{\lambda \tilde{F}(w^\star)}{\eta + \lambda \tilde{F}(w^\star)}$$

Offered and Accepted Wages

- Optimal decision rule $\rightarrow$ truncation in the accepted wage distribution

$$g(w) = \frac{f(w)}{\tilde{F}(w^\star)}, w \geq w^\star$$

- This density is well defined: integrates to 1 and non-negative for all $w \geq w^\star$

Likelihood Contributions

- The likelihood of an ongoing unemployment spell is

$$L(t_u, u) = f_u(t_u)p(u) = \lambda \tilde{F}(w^\star) \exp[-\lambda \tilde{F}(w^\star)t_u] \times \frac{\eta}{\eta + \lambda \tilde{F}(w^\star)}$$

- The likelihood of employed and earning a wage w is

$$L(w, e) = \frac{f(w)}{\tilde{F}(w^\star)} \times \frac{\lambda \tilde{F}(w^\star)}{\eta + \lambda \tilde{F}(w^\star)} = \frac{\lambda f(w)}{\eta + \lambda \tilde{F}(w^\star)}$$

Likelihood Function

- The likelihood function for a random sample of N individuals is then

$$L(w_1, .., w_{N_e}, t_1, .., t_{N_u}) = \prod_{i \in e} \left[\frac{\lambda f(w(i))}{\eta + \lambda \tilde{F}(w^\star)} \right] \times \prod_{i \in u} \left[\frac{\eta \lambda \tilde{F}(w^\star) \exp[-\lambda \tilde{F}(w^\star) t_u(i)]}{\eta + \lambda \tilde{F}(w^\star)} \right]$$

- And the associated log-likelihood is

$$LL = -N \ln[\eta + \lambda \tilde{F}(w^\star)] + N \ln \lambda + \sum_{i \in e} \ln[f(w(i))] + N_u \ln[\tilde{F}(w^\star)] + N_u \ln(\eta) - \lambda \tilde{F}(w^\star) \sum_{i \in u} t_u(i)$$

$\Rightarrow$ $w^\star$ is part of the support of F (and hence of LL)

Order Statistics

- Recover the reservation wage as the minimum accepted wage in the data

$$\hat{w}^\star = \min(w_1, ..., w_{N_e})$$

  $\Rightarrow$ Order statistics converge at rate N; problematic in the presence of measurement error

- Maximize log likelihood with respect to λ, η and μ conditional on $\hat{w}^\star$

  $\Rightarrow$ $F(w)$ needs to be recoverable: $F(w|w \geq \hat{w}^\star) = \frac{F(w)-F(w^\star)}{F(w^\star)}, \forall w \geq \hat{w}^\star$

- Plug estimated parameters into equation for $w^\star$ and solve for either b or ρ

  $\Rightarrow$ Usually fix ρ and recover the value of b

Informality, job search, and schooling investments

Bobba, Flabbi and Levy (IER, 2022)

- An equilibrium job search model where
  - $\Rightarrow$ Search frictions generate mobility between formal and informal jobs
  - $\Rightarrow$ Match productivity and bargaining generate overlapping wage distributions
- One long-term "cost of informality": under-investment in education
  - $\Rightarrow$ Same features that create informality may also distort returns to schooling
  - $\Rightarrow$ Trade-off between welfare in the labor market and pre-market investments in human capital

Labor Markets in Latin America

- More than half of the labor force is in the informal sector
  - $\Rightarrow$ Workers not contributing to and not covered by the social security system
  - $\Rightarrow$ Informal employees and (most of the) self-employed gained access to non-contributory social programs
- Neither a segmented or a competitive labor market
  - $\Rightarrow$ Individuals transit back and forth between formal and informal jobs
  - $\Rightarrow$ Wage/productivity distributions overlap
  - $\Rightarrow$ Mix of formality status within the same firm

Data

- Mexico's Labor Force Survey (ENOE) in 2005 (estimation) and 2006 (validation)
  - $\Rightarrow$ Each household is interviewed every three months for a total of five interviews
  - $\Rightarrow$ Nonagricultural, full-time, male, private-sector workers
  - $\Rightarrow$ Secondary-school between the ages of 25 and 55 who reside in urban areas
- Key variables
  - $\Rightarrow$ $w \equiv$ Hourly wages as employee, main job after labor contributions
  - $\Rightarrow$ $y \equiv$ Hourly labor income as self-employed, without paid employees
  - $\Rightarrow$ $f = 1$ if employee is contributing to the social-security fund; $= 0$ otherwise
  - $\Rightarrow$ $h = 1$ if Upper secondary completed $= 0$ if Lower secondary completed

Descriptives: Cross Section (1st quarter of 2005)

Labor Market State:	Unemployed	Formal Employee	Informal Employee	Self-Employed
	High Schooling ($N = 5,044$)			
Proportion (percent)	5.06	58.90	19.77	16.28
Hourly Earnings: Mean		27.16	20.04	23.38
Hourly Earnings: SD		16.28	14.53	16.29
	Low Schooling ($N = 8,570$)			
Proportion (percent)	4.36	56.16	23.40	16.08
Hourly Earnings: Mean		22.31	16.72	21.18
Hourly Earnings: SD		11.01	9.03	13.63

Descriptives: Yearly Transition Rates (balanced panel Q1-Q5, 2005)

High Schooling ($N = 1,330$)				
Labor Market State Q1: $\rightarrow$	Unempl.	Formal Empl.	Informal Empl.	Self-Empl.
Labor Market State Q5: $\downarrow$				
Unemployed	31.58	52.63	15.79	0.00
Formal Employee	1.75	91.43	5.47	1.34
Informal Employee	3.49	21.49	68.86	6.14
Self-Employed	0.00	9.65	13.16	77.19

Low Schooling ($N = 2,329$)				
Labor Market State Q1: $\rightarrow$	Unempl.	Formal Empl.	Informal Empl.	Self-Empl.
Labor Market State Q5: $\downarrow$				
Unemployed	8.82	63.24	26.47	1.47
Formal Employee	2.10	87.80	8.66	1.44
Informal Employee	1.77	27.76	63.39	7.09
Self-Employed	0.44	10.09	16.23	73.25

The Model Environment

- Timing
  - ⇒ Schooling decision
  - ⇒ Searching status decision
  - ⇒ Labor market dynamics
- Labor Market States
  - ⇒ Unemployed
  - ⇒ Self-employed
  - ⇒ Informal Employee
  - ⇒ Formal Employee

Schooling Decision

- Irrevocable decision about schooling level $h \in \{0, 1\}$
- Individual-specific heterogeneity
  - costs $\kappa \sim T(\kappa)$
  - opportunity cost - PDV of participating in LMK as $h = 0$
- Only agents with $\kappa < \kappa^\star(y)$ will acquire $h = 1$
- All labor market parameters are allowed to be schooling-specific

Searching-status Decision

- Irrevocable decision $s \in \{0, 1\}$:
  - $\Rightarrow$ Self-employed ($s = 1$)
  - $\Rightarrow$ Unemployed ($s = 0$)
- Search for a job in both states but receive offers at different rates: $\gamma_h = \psi_h \lambda_h$, $\psi_h \in (0, 1)$
- Self-employment income $y \sim R(y|h)$
- Only agents with $y \geq y^\star(h)$ search while also working as self-employed

Flow Values (Net of Wages) of Formal and Informal Jobs

- Workers
  - $\Rightarrow$ Formal employee $= \beta_{1,h} B_1[w_1(x;y,h)] = \beta_{1,h}[\tau t w_1(x;y,h) + b_1] \rightarrow$ spillovers within and across h
  - $\Rightarrow$ informal employee $= \beta_{0,h} B_0 \rightarrow$ exogenous and varies with policy (more on this later)
- Firms
  - $\Rightarrow$ Match-specific productivity: $x \sim G(x|h)$
  - $\Rightarrow$ One-shot penalty for hiring illegally: $c_h w_0(x;y,h)$

Labor Market Dynamics

State	PDV	Shock	Flow Utility
Workers:			
Unemployed	$U(h)$	λ_h	$\xi_h + \beta_{0,h} B_0$
Self-Employed	$S(y,h)$	γ_h	$y + \beta_{0,h} B_0$
Informal Employee	$E_0[w,y,h]$	η_h, χ_h	$w_0(x;y,h) + \beta_{0,h} B_0$
Formal Employee	$E_1[w,y,h]$	η_h, χ_h	$w_1(x;y,h) + \beta_{1,h} B_1[w_1(x;y,h)]$
Firms:			
Vacancy	$V[h]$	ζ_h	ν_h
Filled Informal Job	$F_0[x,y,h]$	η_h, χ_h	$x - (1+\chi_h c_h) w_0(x;y,h)$
Filled Formal Job	$F_1[x,y,h]$	η_h, χ_h	$x - (1+t) w_1(x;y,h)$

- Matching function determines $\{\lambda_h, \gamma_h, \zeta_h\}$: $m_h = (u_h + \psi_h s_h)^{\iota_h} (v_h)^{1-\iota_h}$

Wage Determination

- Wages are determined upon meeting by bargaining + firm posting formality status f

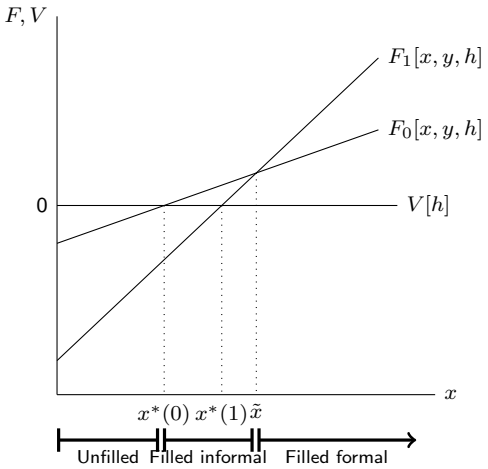
$$\max_{w|f}\{E_f[w,y,h]-Q(y,h)\}^{\alpha_h}\{F_f[x,y,h]-V[h]\}^{(1-\alpha_h)}, \alpha_h \in (0,1)$$

$\Rightarrow Q(y,h) \equiv \max\{S(y,h), U(h)\}$

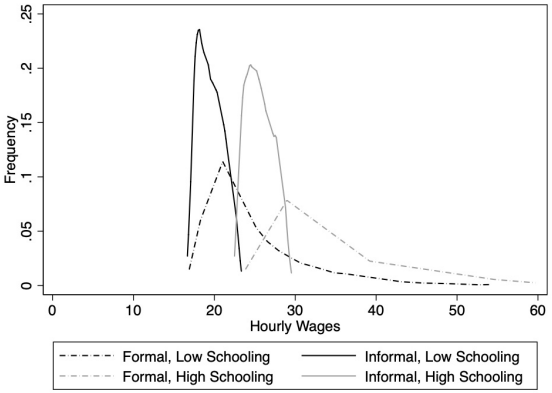
- Equilibrium wage schedules (under free-entry of firms: $V[h]=0$) are:

$$\begin{aligned} w_0(x;y,h) &= \frac{\alpha_h}{1+\chi_h c_h}x + (1-\alpha_h)[\rho Q(y,h) - \beta_{0,h}B_0] \\ w_1(x;y,h) &= \frac{\alpha_h}{1+t}x + \frac{(1-\alpha_h)}{1+\beta_{1,h}\tau t}[\rho Q(y,h) - \beta_{1,h}b_1] \end{aligned}$$

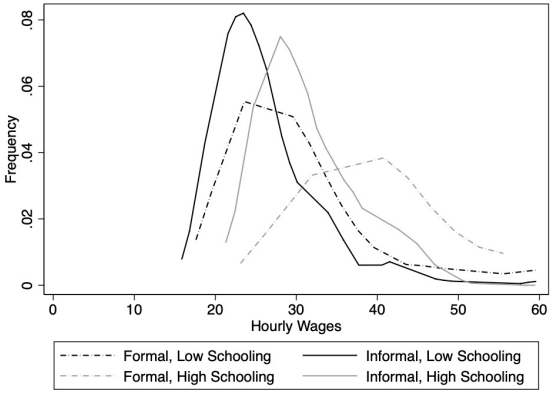
Equilibrium Representation



Model-based Accepted Wage Distributions By Search Status



(a) Outside Option: Unemployment



(b) Outside Option: Self-employment

Empirical Implications

- Main stylized facts of informal labor markets are replicated in equilibrium
  - $\Rightarrow$ A mixture of formal and informal jobs is realized
  - $\Rightarrow$ Formal employees have on average higher wages than informal employees
  - $\Rightarrow$ But their accepted wage distributions overlap
  - $\Rightarrow$ Informal employees and self-employed have different labor market dynamics
  - $\Rightarrow$ Workers transit over time between different formality status

Search-Matching-Bargaining Parameters

- $G(x|h)$: Has to be "recoverable"
  - $\Rightarrow$ We assume lognormal with parameters $\{\mu_{x,h}, \sigma_{x,h}\}$
- $\lambda_h, \gamma_h, \eta_h$: stationarity + optimal decision rules identify mobility rates from
  - $\Rightarrow$ Transitions
  - $\Rightarrow$ Steady state distributions over labor market states
- ρ, ξ_h: Use $Q(y,h)$ to obtain their joint identification
- Nash Bargaining coefficient: $\alpha_1 = \alpha_0 = \alpha$
  - $\Rightarrow$ Use aggregate labor shares $(w_f(x;y,h)/x)$

Self-employment and Schooling Parameters

- $R(y|h)$: Identified by observed self-employment earnings, recoverable primitive distribution

  $\Rightarrow$ We assume lognormal with parameters $\{\mu_{y,h}, \sigma_{y,h}\}$

- $T(\kappa)$: Identified from the proportions of individuals in the two schooling levels

$$\frac{1}{n}\sum_{i=1}^{n} h_i = \int_y T(\kappa^*(y))dR(y|0)$$

  $\Rightarrow$ We assume a negative exponential with parameters δ

Informality Parameters

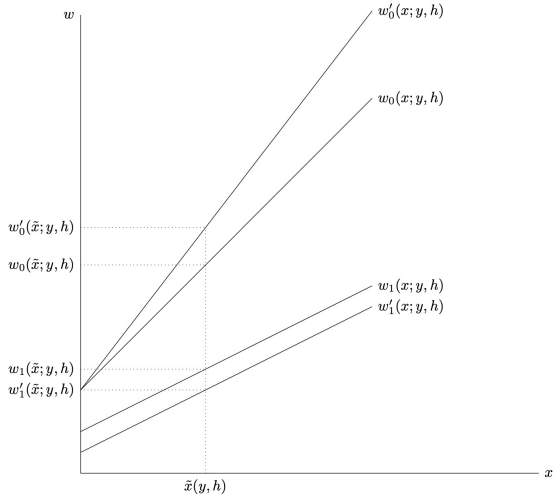
- Different transition rates out of formal jobs and informal jobs identify χ_h
- Overlap between formal and informal accepted wage distributions

$$w_0(\tilde{x}(y,h);y,h) - w_1(\tilde{x}(y,h);y,h) > 0$$

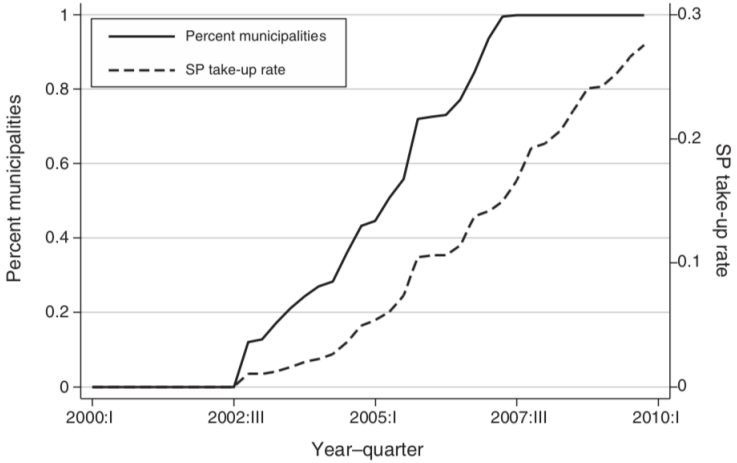
  $\Rightarrow$ β_1 and c_h alter location and overlap of accepted wage distributions

- Variation in y provides additional useful variation to separately identify the parameters

Overlap and Identification of β_1 and c_h



Preference for Informal Extra-wage Benefits (β_0)



Identification of β_0

- Variation in B_0 identifies β_0 if uncorrelated with changes in model primitives
- Labor market outcomes pre-policy (2002) are balanced

	Hourly Wages (log)			Labor Market Proportions			
	Formal	Informal	Self	Formal	Informal	Self	Unempl
SP in 2005 (1=yes)	-0.041	0.048	-0.035	-0.034	0.035	-0.004	0.003
	(0.036)	(0.055)	(0.062)	(0.026)	(0.019)	(0.014)	(0.006)
Complete Sec. (1=yes)	0.218	0.288	0.092	0.061	-0.036	-0.029	0.003
	(0.017)	(0.032)	(0.033)	(0.011)	(0.008)	(0.008)	(0.003)
Number of Obs.	7865	5474	2777	16458	16458	16458	16458

Unobserved Worker Types

- We denote each type with $k \in \{1, 2\}$ and its proportion in the population with π_k

$$x|k = a_k^G x$$
$$y|k = a_k^R y$$
$$\kappa|k = a_k^T \kappa$$

  $\Rightarrow$ Type $k = 1$ normalized to $a_1^T = a_1^R = a_1^G = 1$

  $\Rightarrow$ Type $k = 2$ exhibiting $a_2^T < 1; a_2^R > 1; a_2^G > 1$

- Duration dependence in unemployment identifies these parameters

  $\Rightarrow$ Hazard rates at three and six months for both schooling levels

Estimation: Simulated Method of Moments

- Inner loop to match the following moments for $h \in \{0,1\}$ and SP $\in \{0,1\}$
  - $\Rightarrow$ Shares in each labor market state + yearly transitions ($\lambda_h, \gamma_h, \chi_h$)
  - $\Rightarrow$ Accepted wage distributions of formal and informal employees ($\mu_{x,h}, \sigma_{x,h}, \beta_{1,h}, \beta_{0,h}, c_h$)
  - $\Rightarrow$ Accepted earnings distributions of self-employed ($\mu_{y,h}, \sigma_{y,h}$)
  - $\Rightarrow$ Hazard rates out of unemployment at 3 and 6 months ($a_2^T, a_2^R, a_2^G, \pi_1, \pi_2$)
  - $\Rightarrow$ Aggregate labor Shares ($\alpha_1 = \alpha_2 = \alpha$)

- Outer loop to estimate demand-side parameters using vacancy rates given λ_h, γ_h

$$\begin{aligned} \psi_h &= \frac{\gamma_h}{\lambda_h} \\ \iota_h &= \frac{\ln \omega_h - \ln \lambda_h}{\ln \omega_h}. \end{aligned}$$

Estimates: Returns to Schooling

Ability:	Low $k=1$	High $k=2$
PDV of Labor Market Search:		
$\int_y Q(y,h)dR(y\|h)$	0.309	0.278
Average Accepted Wages:		
F: $E_h\left[w_1 \mid \tilde{x}(y,h) \leq x\right]$	0.479	0.435
I: $E_h\left[w_0 \mid x_0^*(y,h) \leq x < \tilde{x}(y,h)\right]$	0.281	0.296
Average Offered Wages:		
F: $E_h\left[w_1 \mid y < y^*(h)\right]$	0.213	0.210
F: $E_h\left[w_1 \mid y \geq y^*(h)\right]$	0.213	0.204
I: $E_h\left[w_0 \mid y < y^*(h)\right]$	0.133	0.134
I: $E_h\left[w_0 \mid y \geq y^*(h)\right]$	0.142	0.136

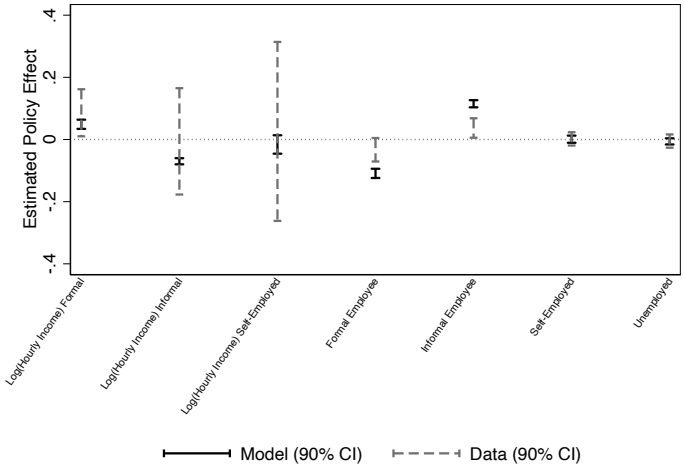
Out-of-Sample Model Validation

- Estimate the effect of $\uparrow B_0$ using SP roll-out in 2006

$$y_{i,q} = \theta d_{m(i),q} + \vartheta h_i + \varrho_{m(i)} + \varphi_q + \epsilon_{i,q}$$

- Predict change in LMK outcomes with B_0^{2006} using estimated model
- Estimate TWFE specifications on both actual and simulated data

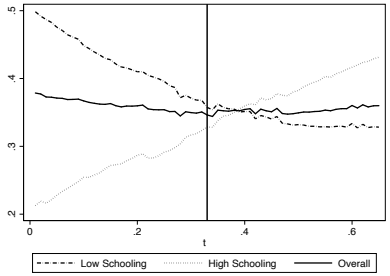
Out-of-Sample Model Validation



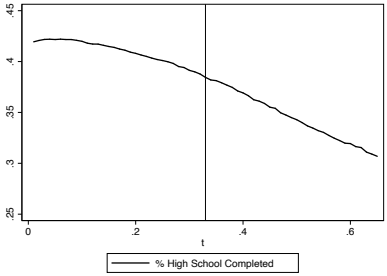
Counterfactual 1: The Equilibrium Effects of Informality

Model:	Firms can only offer a formal contract			
Specifications:	Baseline Model	Exogenous Schooling	Exogenous Contact Rates	Hosios-like Condition ($\alpha = \iota$)
Flow Welfare:				
Total	-0.0596	-0.0750	-0.0020	0.0478
Workers	-0.0460	-0.0599	0.0166	0.0570
Firms	-0.2821	-0.3219	-0.3055	-0.1589
Labor Market Proportions:				
Unemployed	0.0213	0.0636	0.0019	-0.0459
Self-employed	0.3353	0.3526	0.3625	0.2329
Formal Employees	0.0275	-0.0146	-0.0376	0.0076
Schooling Outcomes:				
% HS Completed	0.1029	–	0.0781	0.1501
% High Ability in HS	0.0538	–	0.0569	0.0628

Counterfactual 2: Changes in Payroll Tax Rate (t)



(c) Informality



(d) Schooling

- Composition effect over schooling/ability explain no impact on informality
- Balanced-budget policy with $\tau = 0 \rightarrow$ 10% increase in high-school completion

Main Takeways from the Estimated Model

- Returns to schooling are substantial
- Informality is welfare improving but:
  - $\Rightarrow$ Significantly more so for firms than workers
  - $\Rightarrow$ Reduces human capital accumulation (hold-up problem)
- Payroll tax rate has a non-intuitive impact on equilibrium outcomes
  - $\Rightarrow$ Informality rate not a good indicator for policy
  - $\Rightarrow$ Redistributive forces within the formal sector are key

DID $\Leftrightarrow$ Economic Model

- Relevant institutional features are included in the model in a tractable way
- These parameters are hard to separately identify using model-based restrictions
- The staggered roll-out of the policy provides an additional source variation
  - $\Rightarrow$ Identify the (average) valuation of non-contributory benefits
  - $\Rightarrow$ Validate the model predictions on a different time period by simulating one-step ahead